

such as Miikkulainen *et al.* [41] and the recent updates by Popov *et al.* [52] further map the evolution of ALD chemistries and temperature windows, while the *Nature Reviews Methods Primers* article by Kessels *et al.* captures the current breadth of ALD applications [36].

To demonstrate machine-actionable modeling in a concrete ALD setting, we selected three recent review papers. The main criterion for selection was that each paper contains at least one substantive review table that compares parameters or performance measures across multiple studies and includes a dedicated “Ref.” or “References” column. This aligns naturally with the ORKG publishing model, where table rows are linked back to the original papers, as illustrated in Figure 2. The decision to select the relevant papers to model was made by the first author of this work. In the following subsections, we convert these tables into ORKG Comparisons and query them using two types of example questions: *Easy Queries*, which operate on a single table with simple filters, and *Complex Queries*, which combine several properties, join multiple tables, or compute derived quantities.

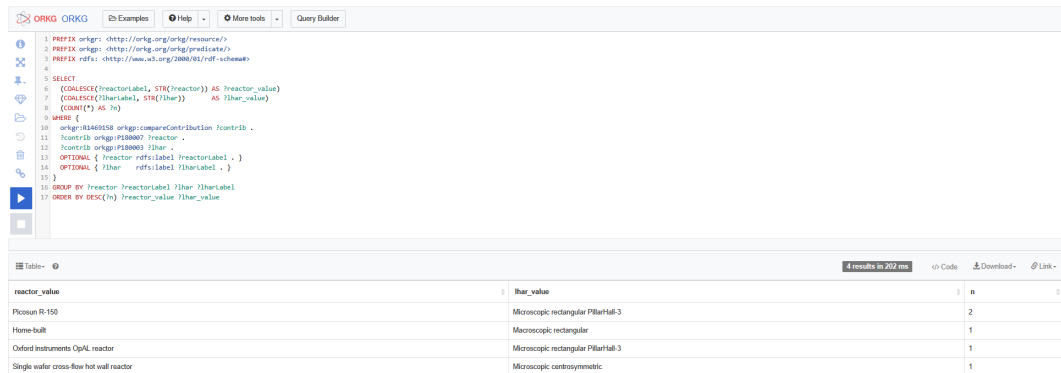


Figure 5: Screenshot of the visual SPARQL editor provided on the ORKG platform (<https://orkg.org/sparql/>), showing the execution of the Q.1 for Table 2 from the research article “Saturation profile based conformality analysis for atomic layer deposition: aluminum oxide in lateral high-aspect-ratio channels” [66], modeled in the ORKG as the machine-actionable Comparison R1469158. The query retrieves co-occurrences of reactor types and LHAR structures across studies, illustrating how different experimental setups pair deposition systems with test geometries. The persistent query URL can be reproduced at <https://tinyurl.com/lhr-reactor>.

3.1 Paper — Saturation profile based conformality analysis for ALD in LHAR channels

We begin with the paper by Yim *et al.* [66], whose Table 2 is modeled in the ORKG as the machine-actionable Comparison <https://orkg.org/comparisons/R1469158>. The table reports sticking coefficients for TMA (c_{TMA}) and H_2O ($c_{\text{H}_2\text{O}}$) during Al_2O_3 ALD in lateral high-aspect-ratio (LHAR) structures, together with process conditions such as deposition temperature, LHAR geometry, number of ALD cycles, and reactor configuration. Because the table includes a dedicated “Reference” column linking each row to its source article, it fits naturally into the ORKG comparison model, where every data row is traceable to an underlying publication.

In the ORKG, the Table 2 comparison <https://orkg.org/comparisons/R1469158> is published as a citable dataset [67] with its own DOI, 10.48366/R1469158. As illustrated earlier in panel 2 of Figure 1, the structure differs slightly from a conventional spreadsheet: each column is an ORKG *contribution* (one device or experimental configuration from a specific article), and each row corresponds to a property (e.g., reactor type, LHAR geometry, c_{TMA}) whose values are compared across contributions. In other words, the original review table is “turned on its side,” but all the same numbers and labels are preserved and each data point remains linked to its source paper.

For the scientific community, this comparison-as-dataset publication model within the ORKG digital library offers two concrete incentives beyond a static CSV upload to a repository such as Zenodo. First, a carefully curated review table becomes a standalone research object with its own DOI that can be cited independently of the article, making the underlying quantitative synthesis more visible and reusable in later work, benchmark studies, or modeling efforts. Second, the comparison is a living resource: new findings are added as *additional columns* (new contributions from subsequent studies) while the shared set of rows (properties) remains stable. Updated versions can be republished

under new DOIs, and a versioning system in the ORKG backend preserves all earlier snapshots. Conceptually, this is similar to a Wikipedia-style update model—anyone can extend the comparison as new ALD results appear—but with the added benefit that each published state is frozen, timestamped, and citable. In fast-evolving areas such as LHAR conformality studies, as an example, this allows the community to maintain an up-to-date, machine-actionable comparison without losing track of the state of knowledge at earlier points in time.

Table 1: Answer table for Q.2 over Table 2 of Yim et al. [66], modeled in the ORKG as Comparison R1469158, listing contributions reporting c_{TMA} values at 300 °C in PillarHall-3 geometries.

Contrib.	Research problem	Temp. [°C]	LHAR Type	c_{TMA}
R1469105	<i>Variation of precursor sticking coefficients in ALD on high-aspect-ratio structures</i>	300	Microscopic rectangular PillarHall-3	0.00572
R1469111	<i>Variation of precursor sticking coefficients in ALD on high-aspect-ratio structures</i>	300	Microscopic rectangular PillarHall-3	0.00400

Easy Query. The first query we introduce in this work is the following natural-language question: “Which combinations of reactor type and LHAR structure are reported across the studies, and how often does each occur?” Once Table 2 is represented as an ORKG Comparison, a question of this kind can be answered programmatically rather than by manually scanning multiple PDF tables. We therefore implement it as a SPARQL query—the standard declarative query language for knowledge graphs, playing a role similar to SQL for relational databases—because it allows the structured table data to be searched, filtered, and aggregated in a transparent and reproducible way. In the specific query shown in Figure 5, the PREFIX declarations at the top define shorthand names for ORKG resources (`orkgr:`), properties (`orkgp:`), and standard RDF terms (`rdfs:`); the SELECT clause specifies which variables appear as columns in the result table (here: reactor type, LHAR structure, and the count of their co-occurrences); and the WHERE block describes the pattern of links in the ORKG knowledge graph that connects each comparison contribution to its reactor and LHAR features. OPTIONAL clauses request human-readable labels when available, while GROUP BY and ORDER BY aggregate and sort the output. In this way, SPARQL provides a compact, readable way to formulate “show me all reactor–geometry combinations and how often they appear” directly over the machine-actionable representation of the review table.

A natural caveat for the reader may be: “How to obtain a SPARQL query for one’s natural-language question?” There are two practical routes. First, one can write SPARQL manually by becoming familiar with these basic constructs; for readers used to SQL, the analogies above offer an accessible starting point, and the ORKG visual editor further lowers the entry barrier by exposing queries and their results side by side. Second, modern large language models (LLMs) are increasingly effective at translating natural-language questions into valid SPARQL queries when given the relevant property identifiers [64, 17]; in particular, Avila *et al.* [8] explicitly evaluate ChatGPT on text-to-SPARQL generation. In practice, a user can phrase their question in plain English, ask an LLM such as ChatGPT to propose a SPARQL query for the ORKG comparison of interest, and then run or adjust this query in the ORKG editor. This workflow substantially lowers the barrier for interacting with machine-actionable scientific data while preserving full transparency and reproducibility.

The SPARQL implementation of our first question can be rerun via <https://tinyurl.com/1hr-reactor>. Figure 5 displays this query in the visual SPARQL editor on the ORKG platform together with its result table, showing how all reactor–LHAR co-occurrences are extracted directly from the machine-actionable representation of Table 2.

Complex Query. The second query we introduce in this work, and the first *Complex Query*, is the following constrained question: “At 300 °C in PillarHall-3 structures, what c_{TMA} values are reported across studies?” Throughout this paper, we refer to queries as *Complex* when they go beyond simple single-property filters—by combining several properties in the same condition set, joining multiple comparisons, or computing derived quantities. In this first example, the query remains on a single comparison but simultaneously restricts three explicit properties in the ORKG knowledge graph: temperature (P37561), LHAR type (P180003), and the sticking coefficient for TMA, c_{TMA} (P180008); it also optionally records the associated research problem via P32. We implement this question as a SPARQL query that can be rerun via <https://tinyurl.com/pillarhall13-ctma>

over the same ORKG Comparison <https://orkg.org/comparisons/R1469158>. The resulting answers, summarized in Table 1, reproduce the relevant rows from the original table in a fully machine-queryable form.

Together, these two questions show that a conventional ALD review table on LHAR conformality can be turned into a structured ORKG Comparison and queried deterministically using SPARQL, without changing the underlying experimental data. We emphasize that these queries are methodological demonstrators rather than endorsements of any specific test structure, reactor technology, or commercial implementation. As noted above, the underlying review papers were chosen by the first author via an independent literature search based on predefined inclusion criteria (presence of substantive multi-row review tables with a dedicated “Ref.”/“References” column and sufficient citation impact), before any of their authors were invited as co-authors on this article; the authors of the original reviews therefore did not influence the choice of example tables.

3.2 Paper – Atomic layer deposition on particulate materials from 1988 through 2023: A quantitative review of technologies, materials and applications

As our second review paper for ALD, we select the comprehensive quantitative review by Piechulla *et al.* (2025) [51]. This article synthesizes more than three decades of research on ALD applied to particulate materials, systematically analyzing 799 publications to map reactor technologies, coating materials, reactants, supports, and process conditions. The study’s quantitative approach aligns well with our goal of representing structured scientific knowledge in the ORKG, as it captures recurring experimental parameters and cross-references them across a large body of literature.

From this paper, we model its Tables 3 and 4, both of which include a dedicated *Reference* column linking reported data to source publications—making them particularly suitable for ORKG comparison modeling and traceable, queryable data integration.

3.2.1 Machine-actionable modeling of Table 3

Table 3 in the review by Piechulla *et al.* (2025) [51] summarizes ALD coating studies on phosphor materials. It lists, for each study, the phosphor host (e.g., BaMgAl₁₀O₁₇:Eu²⁺, Sr[LiAl₃N₄]:Eu²⁺, Zn₂SiO₄:Mn²⁺), the ALD coating material (such as SiO₂, Al₂O₃, TiO₂, MgO, ZnO, TiN), the precursor scheme (e.g., TEOS/H₂O, TMA/H₂O, TiCl₄/H₂O, DEZ/H₂O), coating thickness (typically 2–50 nm or given in cycles), and the corresponding reference.

We model this table as the ORKG Comparison “*Comparative Studies of ALD Coatings on Phosphor Materials*” at <https://orkg.org/comparisons/R1469383> [48]. In this machine-actionable form, each entry is linked to its source publication and all fields are represented as explicit properties, so that structured questions—such as “Which phosphors were coated with SiO₂?” or “Which precursor schemes were used for Al₂O₃ coatings on Eu²⁺-activated hosts?”—can be answered directly via SPARQL queries rather than by re-reading the underlying PDF table.

Table 2: Answer table for Q.3 over Table 3 of Piechulla *et al.* [51], modeled in the ORKG as Comparison R1469383.

Contribution	Phosphor	Coating
R1469163	BaMgAl ₁₀ O ₁₇ :Eu ²⁺	SiO ₂
R1469173	BaMgAl ₁₀ O ₁₇ :Eu ²⁺	SiO ₂
R1469216	Sr[LiAl ₃ N ₄]:Eu ²⁺	SiO ₂

Easy Query. In our question-answering dataset, the third question is: “Which phosphors were coated with SiO₂?” over the ORKG comparison of Table 3. Once the table is modeled in the ORKG, this question can be answered programmatically rather than by re-reading the original review. We implement it as a SPARQL query that can be rerun via <https://tinyurl.com/Phosphor-SiO2-ALD>. In contrast to traditional QA tasks that return free-text responses, our system produces a synthesized, evidence-traceable answer table: Table 2 lists all phosphor materials in the comparison reported with SiO₂ ALD coatings, each explicitly linked to its original contribution. This tabular output mitigates long, potentially redundant narratives and supports transparent verification of the underlying data.